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Empirically derived evaluation requirements for responsible deployments of AI in safety-critical settings

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Processes to assure the safe, effective, and responsible deployment of artificial intelligence (AI) in safety-critical settings are urgently needed. Here we show a procedure to empirically evaluate the impacts of AI augmentation as a basis for responsible deployment. We evaluated three augmentative AI technologies nurses used to recognize imminent patient emergencies, including combinations of AI recommendations and explanations. The evaluation involved 450 nursing students and 12 licensed nurses assessing 10 historical patient cases. With each technology, nurses' performance was both improved and degraded when the AI algorithm was most correct and misleading, respectively. Our findings caution that AI capabilities alone do not guarantee a safe and effective joint human-AI system. We propose two minimum requirements for evaluating AI in safety-critical settings: (1) empirically measure the performance of people and AI together and (2) examine a range of challenging cases which produce a range of strong, mediocre, and poor AI performance.

The potential impacts of artificial intelligence (AI), machine learning (ML), and other forms of advanced automation have sparked polarizing debates among experts[1], practitioners[2], and the general public[3]. Promising demonstrations of AI capabilities[4] have motivated the increasing application of AI in safety-critical settings, like digital medicine[5]. In contrast, recognition of AI errors[6], harms[7], and failed implementations[8] have amplified calls for responsible deployment[9]. Emerging ethical principles for responsible AI practices[10] have reflected a growing consensus that AI should assist or augment rather than replace people in safety-critical settings[5,11]. This has prompted an increased emphasis on human-AI interactions when developing augmentative AI technologies[12], including readily explainable AI outputs[13]. However, both the impacts of AI augmentation[5,14] and methods of evaluation[15,16] have been highly variable. Therefore, it remains unclear whether deployments of augmentative AI for digital medicine will be helpful or harmful across a range of difficult scenarios.

Historically, widespread deployments of digital technologies in healthcare have often introduced new risks, many of which were only recognized after contributing to patient harm. The digitization of medication ordering intended to reduce errors was found to facilitate new kinds of medication errors[17]. Clinical alarms intended to detect patient hazards were identified as hazardous themselves[18,19]. Algorithms intended to alleviate work or improve health outcomes were observed to exhibit poor performance in practice[20], impose additional workload for clinicians[21], and propagate harmful biases and inequities[22]. Recent evaluations have suggested augmentative AI may pose similar risks and introduce new vulnerabilities. For example, studies have shown that it can be difficult for expert clinicians to recognize and recover from erroneous AI recommendations[11,23,24].

Explanations of AI recommendations are expected to mitigate the risks of AI errors[25]; however, evaluations of this claim have been limited. First, empirical evaluations of AI explanations which involve domain practitioners are rare[15]. Second, few studies examine how AI explanations impact the performance of the joint human-AI system[16]. Third, most evaluations do not directly examine whether AI explanations help people recognize and recover from AI errors[26]. Finally, little research has examined the effectiveness of AI explanations in time-sensitive situations[27]. With some exceptions[28,29], evaluations of AI augmentation in healthcare which address these limitations are surprisingly rare.

In the present study, we evaluated the impacts of three augmentative AI technologies nurses used to complete a difficult, time-sensitive, and safety-critical task: early recognition of imminent patient deterioration[30]. We utilized a predictive algorithm[31] with performance that was substantially better than clinical alarm technologies[18], comparable to similar clinical early warning scores[32] and prediction algorithms[33], and sufficiently reliable

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according to proposed benchmarks[34]. We also utilized a patient data interface[35] and explainable AI technique[36] which were informed by guidelines for human-computer[37] and human-AI[12,38] interaction. Similar deployments of AI have been linked to large-scale population benefits[39]. Despite these favorable benchmarks, our findings revealed a combination of helpful and harmful impacts of AI augmentation corresponding to cases in which the AI algorithm was strongly correct and erroneous, respectively. These impacts were not significantly improved by AI explanations. Our findings suggest that AI augmentation can both improve and degrade joint human-AI performance, the impacts of AI augmentation strongly depend on problem difficulty, and explainable AI may be insufficient to mitigate these risks.

From these empirical findings, we derive two minimum evaluation requirements for the responsible deployment of augmentative AI in safety-critical settings: (1) empirically measure the performance of people and AI together and (2) examine a range of challenging cases which produce a range of strong, mediocre, and poor AI performance. Our findings revealed rare but potentially severe consequences of misleading AI recommendations which can be overlooked or underappreciated if evaluation procedures are not structured in this way. Understanding these potential risks before people are harmed is a fundamental responsibility of safety[40]. Therefore, we propose these two requirements as a basis for evaluating AI augmentation in safety-critical settings, assuring safe and effective deployments, and establishing standards of responsible AI practice[41].

Results

We conducted a large-scale repeated measures study involving 450 nursing students and 12 licensed nurses. Each participant analyzed a randomized sequence of 10 historical patient cases with a randomized sequence of 2 × 2 experimental conditions: with and without both AI recommendations and explanations. In all conditions, participants were provided with a data visualization showing the patient demographics, vital sign data, and laboratory results which nurses use to anticipate imminent deterioration[30] (see Supplementary Fig. 1). If provided, AI recommendations showed the predicted probability (0–100%) of an emergency event occurring in the next five minutes[31] (see Supplementary Fig. 2) and AI explanations showed a visual depiction of which data strongly contributed to the recommendation[36] (see Supplementary Fig. 3). Figure 1 shows an example patient case with both AI recommendations and explanations shown. The experimental setup represented how this envisioned technology could facilitate remote patient monitoring. Prior testing with practicing nurses produced promising evidence that the data visualization[35] and AI explanation[36] techniques aided nurses in remotely monitoring patients for early signs of deterioration and recognizing some forms of AI errors.

Participants were instructed to work as quickly as possible and asked to report their concern for each patient on a continuous scale from 0 to 10 with integer reference markers. In half of the historical patient cases, an emergency event occurred five minutes after the most recent data shown to participants. Cases were selected to sample a wide range of AI recommendations corresponding to a wide range of AI performance, including predicted probabilities ranging from 15 to 93% for the five emergency cases and ranging from 12 to 91% for the five non-emergency cases. The predicted probabilities in the set of emergency cases approximately matched the predicted probabilities in the set of non-emergency cases. Participants received no feedback during the study to mitigate the potential for carry-over effects. A summary of results in each case (Supplementary Table 1) and the distribution of nurses' responses by participant group (Supplementary Fig. 4), trial number (Supplementary Fig. 5), experimental condition (Supplementary Fig. 6), and changes in experimental condition between trials (Supplementary Figs. 7, 8) are included in the supplementary materials.

Complementarity of nurse and AI performance

To assess the potential for complementary nurse and AI performance[26], we compared nurses' concern without AI augmentation to the corresponding AI recommendations in each case. To facilitate comparison, AI

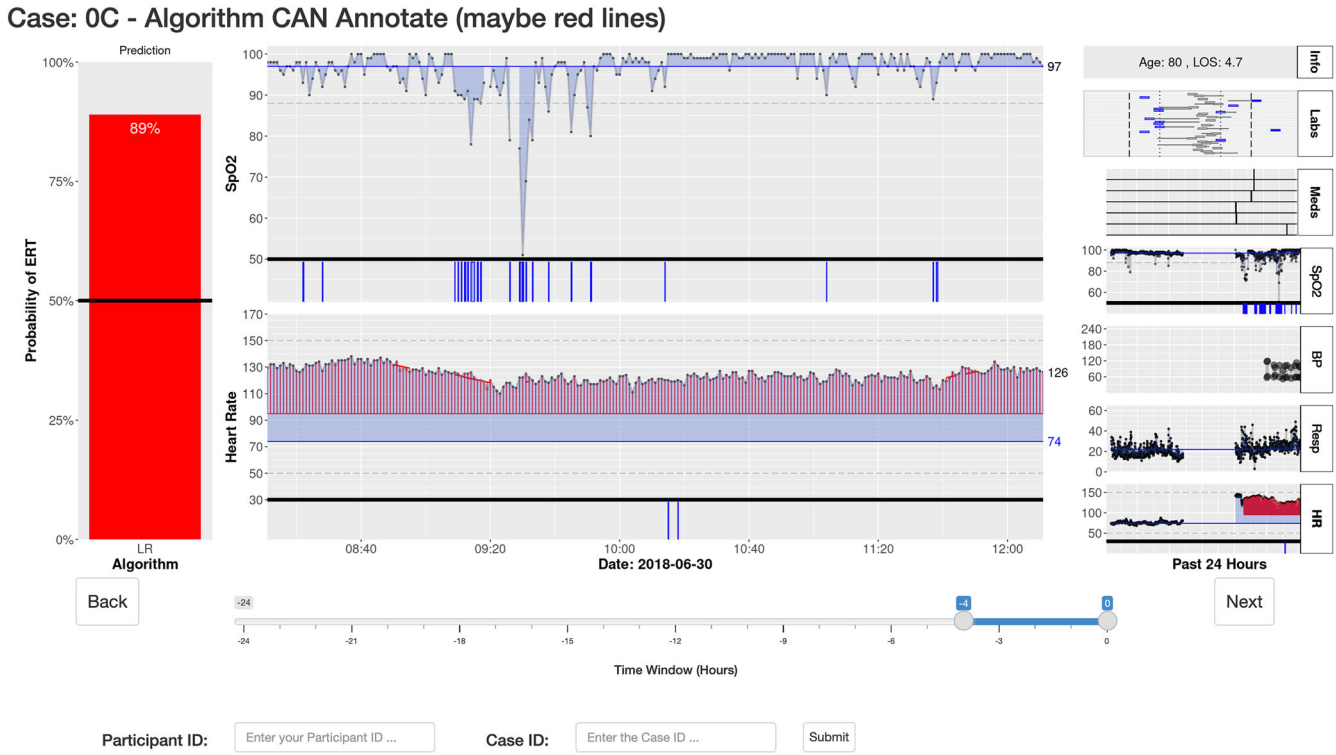

Fig. 1 | Example patient case with AI recommendations and explanations. AI recommendations appeared as a salient red bar at the left of the patient visualization, corresponding to the predicted probability (0–100%) of an emergency event occurring in the next five minutes. AI explanations appeared as a set of red data-driven annotations which visually highlighted the data most strongly contributing to the AI recommendation. All other elements of the patient data display were included in all experimental conditions.

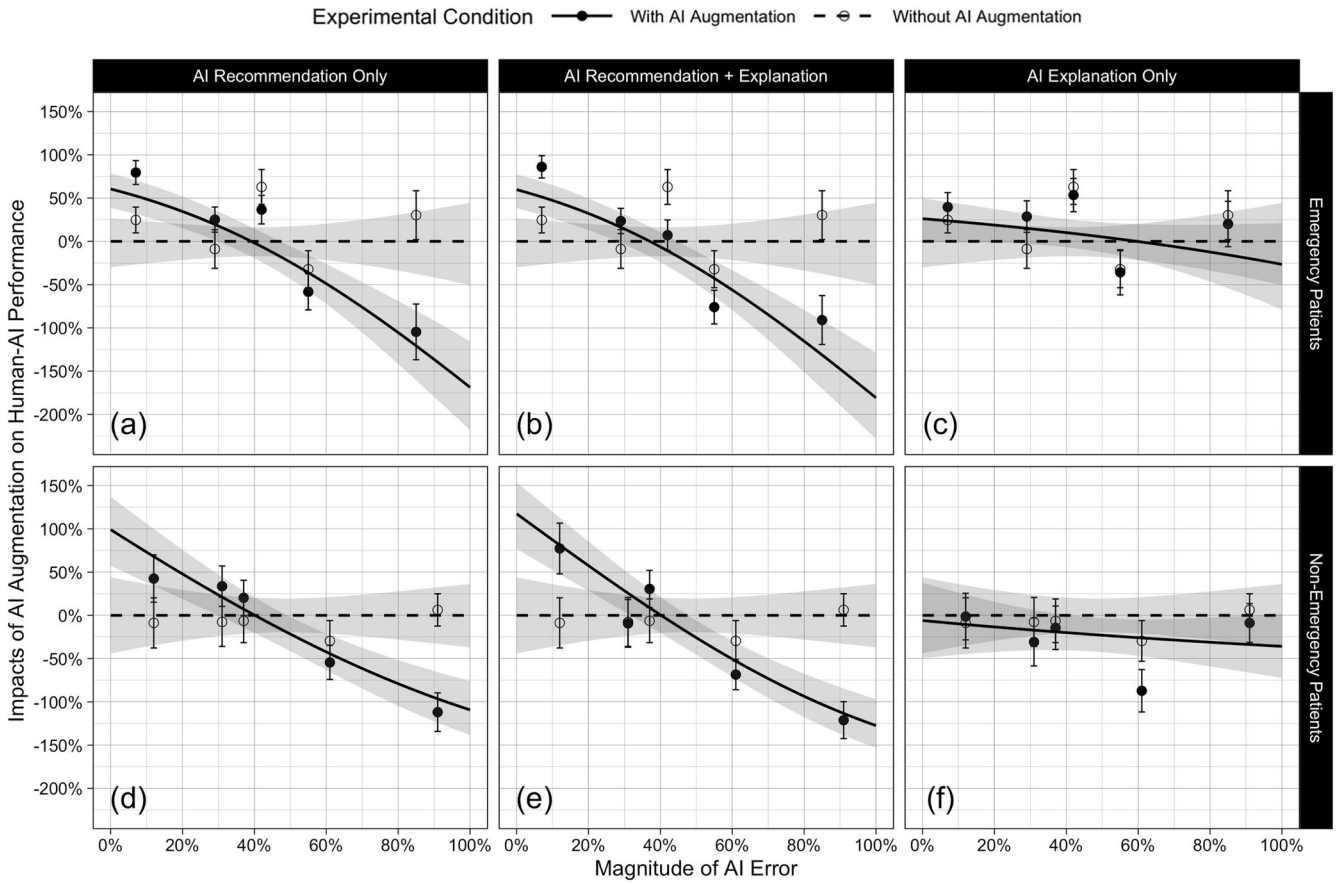

Fig. 2 | Impacts of AI augmentation on human-AI performance. The magnitude of AI error was calculated as the absolute difference between AI recommendations and ground truth (0% for non-emergency patients or 100% for emergency patients). The impact of AI augmentation in the three AI-augmented conditions was calculated as the percentage difference between nurses' concern with and without AI augmentation divided by the extent to which nurses differentiated emergency from non-emergency patients without AI augmentation. Positive and negative values correspond to changes in nurses' concern which increased and decreased nurses' differentiation of patient types, respectively. For emergency patients, results are separated by experimental condition: AI recommendation only (**a**), AI recommendation and explanation (**b**), and AI explanation only (**c**). For non emergency patients, results are similarly separated by experimental condition: AI recommendation only (**d**), AI recommendation and explanation (**e**), and AI explanation only (**f**). Each line represents the estimated marginal means from the generalized linear mixed effects model for nurses' reported concern with 95% confidence intervals. Circles represent the mean of nurses' concern in each case calculated from the unmodeled data, transformed with the same percentage difference calculation. Error bars correspond to 95% confidence intervals. Solid lines and closed circles represent the results of nurses using one of the three AI-augmented experimental conditions. Dashed lines and open circles represent the results of nurses using the baseline experimental condition without AI augmentation.

recommendations (0–100%) were transformed to the same scale as nurses' concern (0–10). We used one-sample two-sided t tests with a Bonferroni correction for 10 comparisons to adjust the level of significance ($\alpha = 0.005$). We considered higher concern for emergency patients and lower concern for non-emergency patients to be indicators of better performance. We found neither nurses nor the AI algorithm were universally superior to the other in all cases (Supplementary Fig. 9). Nurses without AI augmentation reported concern that was significantly better than the AI algorithm in 4 cases (3 emergency, 1 non-emergency), significantly worse than the AI algorithm in 4 cases (1 emergency, 3 non-emergency), and not significantly different than the AI algorithm in 2 cases (1 emergency, 1 non-emergency).

Impacts of AI augmentation

To understand the effects of AI augmentation on nurses' concern, we used a generalized linear mixed effects model. We included all possible interactions between patient type (emergency vs. non-emergency), AI recommendation (0–100%), and 2×2 experimental condition as fixed effects. We also included crossed random intercepts to account for both repeated measures from participants and any additional effects of each patient case[42]. Furthermore, we investigated additional terms and interactions to assess the effects of nurses' experience and potential carry-over effects resulting from the progression of trials or experimental conditions. We found participants working without AI recommendations reported progressively lower concern for both emergency (7.19 vs. 6.60, odds ratio = 1.32, 95% CI [1.16, 1.48], $z = 4.56$, $p < 0.001$) and non-emergency (5.47 vs. 4.77, odds ratio = 1.32, 95% CI [1.16, 1.48], $z = 4.56$, $p < 0.001$) patients from the first to last trial. Since this effect represented a change in baseline concern rather than an improvement in differentiating patient types, we report the average effect for the remaining results. We did not find experience or changes in experimental condition to be strong predictors of nurses' concern.

Figure 2 shows the impacts of AI augmentation on human-AI performance relative to the magnitude of AI error for each of the three AI-augmented experimental conditions. AI error was calculated as the absolute difference between the AI recommendation and ground truth (0% for non-emergency patients or 100% for emergency patients). Without AI augmentation, nurses' concern was not significantly correlated with AI recommendations and differentiated emergency from non-emergency patients by an average of 19.5% of the scale (6.84 vs. 4.89, odds ratio = 2.27, 95% CI [1.78, 2.76], $z = 7.39$, $p < 0.001$). The impact of AI augmentation was calculated as a percentage difference relative to this baseline. Differences between nurses' concern with and without AI augmentation were divided by the extent to which nurses differentiated emergency from non-emergency

patients without AI augmentation. Emergency and non-emergency patient classifications were used to characterize the impacts of AI augmentation as positive or negative depending on whether they contributed to increasing or decreasing nurses' differentiation of patient types, respectively.

We found the performance of the joint human-AI system was both helpfully and harmfully impacted by AI augmentation, depending on the correctness of the AI algorithm. Compared to nurses working without AI augmentation, nurses working with AI recommendations performed better on both emergency (+52.8%, odds ratio = 1.86, 95% CI [1.49, 2.23], z = 6.06, $p < 0.001$) and non-emergency (+67.4%, odds ratio = 1.62, 95% CI [1.35, 1.89], z = 5.59, $p < 0.001$) cases when the AI algorithm was most correct, but worse on both emergency (−120.5%, odds ratio = 0.414, 95% CI [0.327, 0.501], z = −8.25, $p < 0.001$) and non-emergency (−96.3%, odds ratio = 0.418, 95% CI [0.333, 0.503], z = −8.41, $p < 0.001$) cases when the AI algorithm was most erroneous or misleading. When AI recommendations were most misleading, nurses' concern was not significantly different from their concern for the *opposite* category of patients without AI augmentation. The addition of AI explanations did not significantly alter results when AI recommendations were also shown. When only AI explanations were shown, nurses performed better on emergency cases when the AI algorithm was most correct (+23.9%, odds ratio = 1.29, 95% CI [1.05, 1.53], z = 2.64, $p = 0.008$), worse on non-emergency cases when the AI algorithm was most misleading (−33.6%, odds ratio = 0.746, 95% CI [0.599, 0.893], z = −2.91, $p = 0.004$), and not significantly different on either emergency cases when the AI algorithm was most misleading or non-emergency cases when the AI algorithm was most correct. Notably, the visual salience of the AI explanations was correlated with the urgency of the recommendations[36], so these results reflect an increasing impact with increasing visual salience.

Discussion

Despite the potential for complementary human and AI performance, we instead found a combination of helpful and harmful impacts of AI augmentation which corresponded to the difficulty of problem for the AI algorithm. From our testing set, we identified several cases in which nurse trainees alone performed both better and worse than the AI algorithm alone. From these separate evaluations, nurses and AI appeared to have complementary strengths and weaknesses with the potential for each to compensate for the shortcomings of the other[26]. However, the three forms of AI augmentation we evaluated did not actualize these potential benefits. We found nurses' performance was improved by the most correct AI recommendations but degraded by the most misleading AI recommendations. The magnitude of maximum degradation was nearly *twice* the magnitude of maximum improvement. When AI recommendations were most misleading, nurses' reports of concern for emergency patients were indistinguishable from non-emergency patients, and vice versa. We observed a similar effect of lesser magnitude when nurses were presented with only AI explanations. These results suggest all forms of AI augmentation significantly influenced nurses' perceptions of patient risk.

Our findings suggest significant interactions between nurses and AI interfered with the potential complementarity of the two. With favorable benchmarks for AI performance[33,34], explanations[38], and interactions[12,37], it was reasonable to believe our augmentative AI technology was capable enough to be deployable. However, the strong influence of AI recommendations appeared to propagate AI vulnerabilities and induce dangerous confusions in cases which were otherwise routine for nurses. Our findings reinforce that the *interactions* between people and augmentative AI technologies can strongly impact performance, undermine apparent redundancies, and even introduce new risks[11,23,24,29,43]. Due to these interactions, optimizing the performance of AI algorithms alone does not necessarily optimize the performance of the joint human-AI system[44,45]. Instead, AI-based algorithms and other forms of technology augmentation should be conceived, modeled, designed, and evaluated with people as a *joint cognitive system*[46]. The goal of developing augmentative AI technologies should not be to improve AI algorithm performance, but rather to enable people to effectively accomplish cognitive work.

Our findings caution that human supervision may not fully mitigate the risks of all AI errors, even when provided with AI explanations. We found that all forms of AI augmentation significantly degraded human-AI performance when the AI algorithm was most misleading. Poor AI performance strongly and consistently induced poor joint human-AI performance. The addition of AI explanations did not attenuate these effects and could themselves degrade performance. These results are especially surprising because we utilized an interpretable[47] and human-centered[31] algorithm with explanations that were designed to highlight algorithm errors[36]. Our findings reinforce that both AI recommendations and explanations can be persuasive and subject to error[48]. Although explanations can sometimes reveal AI errors, explanations can also reinforce misleading recommendations[44,49] or introduce additional sources of misleading information[28,50]. These risks are not unique to AI algorithms but rather fundamental risks of misleading information[51]. Our findings add to growing concerns about the effectiveness of explainable AI to reliably help people recognize and recover from AI errors[26,29,47,48,52–54].

Although different AI algorithms may produce different results, similar findings[11,23,24,29,43] suggest susceptibility to AI errors might be a common feature of recommendation-centric human-AI architectures[43,46,48]. All AI algorithms will produce a range of strong, mediocre, and poor performance. This range is likely to include some instances where people perform both better and worse than the AI algorithm alone. Improving the AI algorithm will likely reduce the *frequency* of poor performance but it may not reduce the *range* of possible performance or the *consequences* of poor performance. Therefore, improving the AI algorithm does not necessarily lessen the impact of AI errors. On the contrary, prior research has suggested recognizing and recovering from errors might be *more* difficult if the frequency of error is rare[55,56]. Future research should investigate alternative human-AI architectures which might be less susceptible to propagating AI errors[48,57,58].

Our findings accentuate the risks of using aggregate measures (e.g., overall classification accuracy) to evaluate the impacts of AI augmentation in safety-critical settings. Aggregation applies a judgment about the relative value of disparate effects, often equating benefits and harms in ways that are misaligned with the magnitude of consequences[59]. For example, we selected cases which varied in difficulty to produce a wide range of AI performance, intentionally over-sampling cases of mediocre or poor AI performance. These cases revealed a combination of AI impacts that both helped and harmed human-AI performance in different circumstances. If we had representatively sampled from the distribution of AI outputs or computed a weighted average of our results based on the probability of occurrence (see Supplementary Fig. 10), the frequent but smaller benefits from correct AI recommendations might have outweighed the rare but larger costs from misleading AI recommendations. However, this aggregation misrepresents how healthcare practitioners are held accountable. Nurses are responsible for patients individually, not in the aggregate. Major harms are not excused by the accumulation of minor benefits. If AI augmentation produces disparate benefits and harms[11,23,24,29], aggregate evaluations can obscure or underappreciate the risks of infrequent but severe vulnerabilities.

Collectively, our findings and the methods through which they were empirically revealed suggest minimum evaluation requirements for responsible AI practices. Responsible deployments of augmentative AI technologies in safety-critical settings require decision-makers to be well-informed about potential risks before people are harmed[40]. This necessitates conducting pre-deployment evaluations to assess post-deployment risks. The study we conducted is an example of one such evaluation which was able to empirically reveal severe potential risks prior to deployment. Our findings suggest two minimum evaluation requirements for establishing a basis of empirical evidence needed to understand and justify risks of deployment. These requirements were derived from our empirical findings but also have strong theoretical connections to concepts of measurement validity[60]. Effective and trustworthy evaluations of augmentative AI should at least satisfy these two aspects of well-informed and valid assessments.

First, well-informed evaluations should *empirically measure the performance of people and AI together*. In safety-critical settings, a growing consensus acknowledges that responsible deployments of AI-based technologies must assist or augment rather than replace people[5,10,11,13]. However, few evaluations of AI augmentation involve people and even fewer involve domain practitioners[15]. Though recent trends towards increased practitioner involvement are promising[23,28,29,61,62], this has yet to become standard practice in the early evaluation of new AI technologies. Evaluations which measure or compare human and AI performance separately[4] must make assumptions about how the resulting joint human-AI system will perform together. Our findings challenge conventional wisdom that improving AI reliability, explainability, or any other capability of a technology alone will necessarily produce a safe and effective joint human-AI system. Instead, our findings caution that even augmentative AI technologies should not be assumed to always exceed the capabilities of early practitioners with effective tools, improve performance in all situations, require minimal training investments, facilitate effective human supervision, or support recovery from technology errors. In the absence of reliable methods to predict human-AI performance from the characteristics of each separately, our findings reinforce that understanding the risks of deployment requires empirically evaluating people and AI together as a joint cognitive system[46]. This is central to establishing *construct validity*, or the match between a measurement (e.g., pre-deployment evaluation) and target phenomenon (e.g., post-deployment risks)[60].

Second, well-informed evaluations should *examine a range of challenging cases which produce a range of strong, mediocre, and poor AI performance*. Our findings reaffirm that recognizing and recovering from AI errors is a consistent and considerable challenge[11,23,24,29,43]. Evaluations which exclusively consider strong AI performance[26], aggregate across disparate regions of AI performance[59], or otherwise obscure areas of AI weaknesses are incomplete assessments of risk. All AI algorithms will produce a range of performance. If practitioners are expected to recognize and recover from AI errors, the impacts must be examined, even if AI errors are expected to be infrequent. The frequency of errors is often underestimated in foresight[63]. Therefore, evaluating a range of AI performance regardless of frequency is central to establishing *content validity*, or the extent to which a measurement samples the entirety of a construct[60]. If AI augmentation sometimes degrades joint human-AI performance, then organizations seeking to deploy AI technologies face an ethical decision. Although organizations might justify different decisions, failing to consider the range of AI performance is a failure to be well-informed about the potential risks of deployment. In safety-critical settings, underinformed evaluations should not be considered responsible, regardless of AI performance.

If results of an evaluation or other factors lead to retraining or altering the AI algorithm[16], the new human-AI system requires additional evaluation. Improving AI performance alone does not necessarily improve the joint performance of people and AI together[44,45]. While some instances of poor AI performance may uncover opportunities to globally improve the AI algorithm, others may confront hard constraints or trade-offs where improving AI performance in some cases degrades AI performance in other cases. In either case, improving instances of poor AI performance narrows the range of AI performance examined in the evaluation, which violates the second evaluation requirement. Well-informed evaluations must subsequently identify and examine new instances of poor AI performance to broadly cover the new range of possible AI performance. Without examining instances of poor AI performance, evaluations are incomplete assessments of joint human-AI performance[26].

Evaluating human-AI performance relative to the range of AI performance forms the basis for evaluating the *joint capabilities* of human-AI systems beyond the individual capabilities of people or AI technologies alone. These joint capabilities include exploiting strong AI performance, recovering from poor AI performance, and improving the performance of both people and AI in situations which are challenging for both[26,46,64]. Understanding the joint capabilities (or deficiencies) of the human-AI system is critically important for understanding risks of deployment. To date, few (if any) augmentative AI algorithms have demonstrated strictly neutral or positive impacts when evaluated across a range of difficult scenarios[11,23,24,29,43]. The evaluation procedure we outlined in this study empirically revealed situations in which AI augmentation both improved and degraded performance. Our findings illustrate the potential positive and negative impacts of AI augmentation which can be overlooked or underappreciated by evaluations that do not examine the joint capabilities of the human-AI system. We hope the requirements derived from our empirical findings can inform the establishment of minimum standards for assuring responsible deployments of augmentative AI technologies in safety-critical settings.

This study had several limitations. First, we were only able to recruit a small sample of licensed nurses to supplement our large sample of nursing students. Although we did not find experience to be a strong predictor of human-AI performance, this study was powered to detect only large differences between nursing students and licensed nurses. This study was not instrumented to detect smaller differences between these two groups or differences within experience levels of licensed nurses. Though studies have found conventional experience-based factors do not reliably predict human-AI performance[23], future research should continue to investigate the relationships between participant experience, behaviors, interactions with AI algorithms, and other factors which may impact human-AI performance. Second, we did not provide feedback during the study, intentionally limiting opportunities for learning to best estimate the impacts of AI augmentation. Future research should investigate the dynamics of how people interact with augmentative AI technologies over time. However, if the benefits of AI augmentation are not immediately actualized or correlated with domain experience[23], deployments of augmentative AI technologies in safety-critical settings might ironically increase requirements for training and human expertise[65,66]. Third, we could only obtain binary ground truth to characterize patient urgency (emergency vs. non-emergency). For patients where a moderately urgent response would be considered best practice, the impacts of AI augmentation in our results could be over-estimated. Fourth, this study presented a computer-based task for remote monitoring of patients which was not situated within the natural clinical environment. We instrumented this study with several aspects of actual clinical work to realistically simulate remote monitoring. This included using actual patient data, real AI predictions and explanations, realistic cognitive work, and variations in difficulty for both nurses and the AI algorithm. However, additional sources of information and environmental context in clinical environments could alter results. Therefore, our results are more likely to generalize to settings where there is physical distance between nurses and patients (e.g., remote monitoring) rather than direct access to patients (e.g., bedside care). Finally, this study was limited to one algorithm, explanation technique, and interface for one setting, target population, and training program. Our choices were representative of common strategies[13,33,39], but not exhaustive. Alternative AI algorithms and forms of augmentation could produce different results. Our findings reinforce the possibility of varied and surprising impacts of AI augmentation[5,14], which in turn underscores the importance of empirical evaluation.

Methods

AI algorithm, interface, and explanations

To provide nurses with context about each patient, we adapted a patient data interface designed to aid early recognition of patient deterioration[36] (see Supplementary Fig. 1). The interface presented seven data sources which nurses utilize to detect early signs of patient deterioration[30], including: patient demographics (age, length of stay, primary diagnosis, and comorbidities), heart rate, respiratory rate, blood pressure, oxygen saturation, laboratory results, and medications. The interface permitted nurses to directly compare up to 24 h of a patient's heart rate with any of the six other data sources available, with patient demographics as the default comparison. Data representations were designed to align with how clinicians interpret signs of patient deterioration[67]. Prior user testing has produced evidence that these representations support nurses in developing an advantageous

understanding of patients for remote monitoring[35,36]. The display utilized in this study was slightly modified from prior versions to improve usability[68].

We utilized a logistic regression algorithm developed with a human-centered approach to predict imminent patient deterioration[31]. Training data was obtained from telemetry monitors and the electronic health record of a general medical/surgical ward of a large midwestern medical center. The model was trained to classify patients who would and would not require mobilization of a response team to provide emergency care within five minutes. All data included in the patient data display were available for developing the model. The interpretability of the model, consistency and quality of data in clinical environments, and similarity to clinical practice were considered in choosing the type of model, data, and features, respectively. Logistic regression was selected because of its interpretability[47], frequent use in healthcare applications[33,69], and demonstrated success in similar safety-critical deployments[39,70]. Data requirements reflected current data collection practices to maximize the availability of the algorithm and minimize performance discrepancies in clinical settings[21]. Model features were developed to closely align with how nurses recognize signs of deterioration[30,67]. The training dataset included a balanced sample of emergency and non-emergency patients with equal weightings for false positive and false negative classifications. Lasso regression[71] was used to reduce risks of overfitting and improve model interpretability by removing correlated features and creating a simpler model, respectively. Additional details about the training datasets, modeling assumptions, and human-centered considerations are published elsewhere[31,36].

The predictive model was evaluated using k-fold cross-validation. Model performance measures included 75% sensitivity (or recall), 67% specificity, 70% positive predictive value (or precision), 72% negative predictive value, 71% accuracy, 72% F1 score, and 0.77 area under the receiver operating characteristic curve (AUC)[31]. The performance of the model was substantially better than most clinical alarm technologies[18], comparable to similar early warning scores[32] and predictive algorithms[33,72–74], and better than the proposed benchmark for helpful augmentative technologies[34]. Nurses were shown the predicted probability of a patient emergency event (0–100%) directly computed from the logistic function, which provided an estimate of model reliability in similar training cases. The model produced a wide range of predicted probabilities (Supplementary Fig. 10) and was well-calibrated across the range of predictions in the training dataset (Supplementary Fig. 11). For emergency patients, predicted probabilities had a mean of 62.2%, a median of 64%, and an interquartile range from 45% to 80.5%. For non-emergency patients, predicted probabilities had a mean of 36.6%, a median of 32%, and an interquartile range from 21% to 48%. Additional details about model validation and benchmarking are published elsewhere[31].

To help nurses understand and evaluate the appropriateness of the AI recommendations, we utilized a salience-based explainable AI technique[13] which provided a visual explanation of model outputs[36]. This technique goes beyond standard saliency maps, which highlight only the magnitude with which data contribute to model predictions[47], to additionally depict the relationships between individual data, how data are computed into features, and how these features contribute to the model prediction[31]. By revealing more of the structure of the algorithm, this technique was designed to address shortcomings of saliency maps[47,75] and facilitate a richer set of human-AI interactions[12,38,76,77], like conveying when a model is misaligned to a situation[78]. Prior research has produced promising preliminary evidence in support of this technique[36].

Collectively, we characterized the data visualization, predictive algorithm, and explanation technique as an *augmentative* AI technology according to the taxonomy of the American Medical Association[79]. The AI recommendations and explanations suggested clinically meaningful output which would need to be interpreted by a clinician. This AI algorithm and patient data display was an envisioned technology intended to aid remote monitoring of patients for clinical deterioration. Therefore, this study represents an early evaluation conducted to identify potentially impactful technologies prior to further implementation and testing (e.g., clinical trials).

Participants and protocol

Over 2 years, 450 nursing students and a small sample of 12 licensed nurses were recruited for participation in the study. Figure 3 summarizes the demographics of the participant cohort. Nursing students were enrolled in undergraduate or graduate prelicensure programs progressing towards a Bachelor or Master of Science in Nursing. Participants included second-, third-, and fourth-year undergraduate students and first- and second-year graduate students ranging from 18 to 51 years of age and less than 6 months to over 4 years of nursing curriculum completed. Years of nursing education were estimated by the Assistant Dean of Baccalaureate Nursing Programs based on student course enrollment and prelicensure program. Licensed nurses included Registered Nurses (RN) and Licensed Practical Nurses (LPN) ranging from 33 to 48 years of age and 16 months to 25 years of healthcare work experience. All participants were told that the research was focused on improving the patient data display they were using. Students were required to complete the study as an educational activity for course credit and could voluntarily opt to allow their responses to be used for research by completing an informed consent process. We obtained informed consent from all participants in this study.

This study was facilitated online and remotely (via desktop, laptop, or tablet computer). First, participants viewed an approximately 11-min training video which described the study protocol, the patient data display, and the experimental conditions. This training video is publicly available[80]. The training video introduced the AI algorithm by describing the number of patients in the training dataset, how to interpret predicted probabilities, and the features which most strongly contributed to the model[31]. Participants were told that the algorithm was approximately 70% accurate on the training dataset but might perform better or worse than this aggregate for individual patients. Participants were also told that the visual explanations[36] were communicating what data most strongly contributed to the algorithm prediction. The training video additionally described an example of how to interpret these visual explanations. After viewing the training video, participants were provided a practice case and as much time as they needed to familiarize themselves with the display. As participants began the study, they were encouraged to work as quickly as possible and instructed to spend no more than 5 min analyzing each patient case, which corresponded to the prediction horizon of the AI algorithm.

The study utilized a repeated measures or within-subjects design where each participant completed a sequence of 10 historical patient cases with 2×2 experimental conditions: without AI augmentation (Supplementary Fig. 1), with AI recommendations only (Supplementary Fig. 2), with AI explanations only (Supplementary Fig. 3), and with both AI recommendations and explanations (Fig. 1). Both the sequence of patient cases and the sequence of experimental conditions were randomized. In all conditions, participants had access to the same patient data visualization (Supplementary Fig. 1). For each case, participants were asked to report their concern for the patient on a continuous scale from 0 to 10. The scale included labeled markers at each integer to facilitate participants utilizing the entire scale, including both endpoints[81]. Also, the currently selected value was shown (to three decimal places) to help participants precisely distinguish across the scale. Similar visual analog scales have been found valid and reliable in a variety of clinical settings[81–83]. We assumed higher concern would correlate with an increase in care activities associated with effective response to patient deterioration[30], but we did not assume that participants would necessarily have equivalent concern baselines. Participants were not provided feedback about patient outcomes or their performance at any point during the study to mitigate the potential for carry-over effects and best estimate the impacts of AI augmentation.

Data validation and analysis

Prior to analysis, collected data were validated by checking for at least two pieces of evidence (from the data entry form, recorded interactions with the

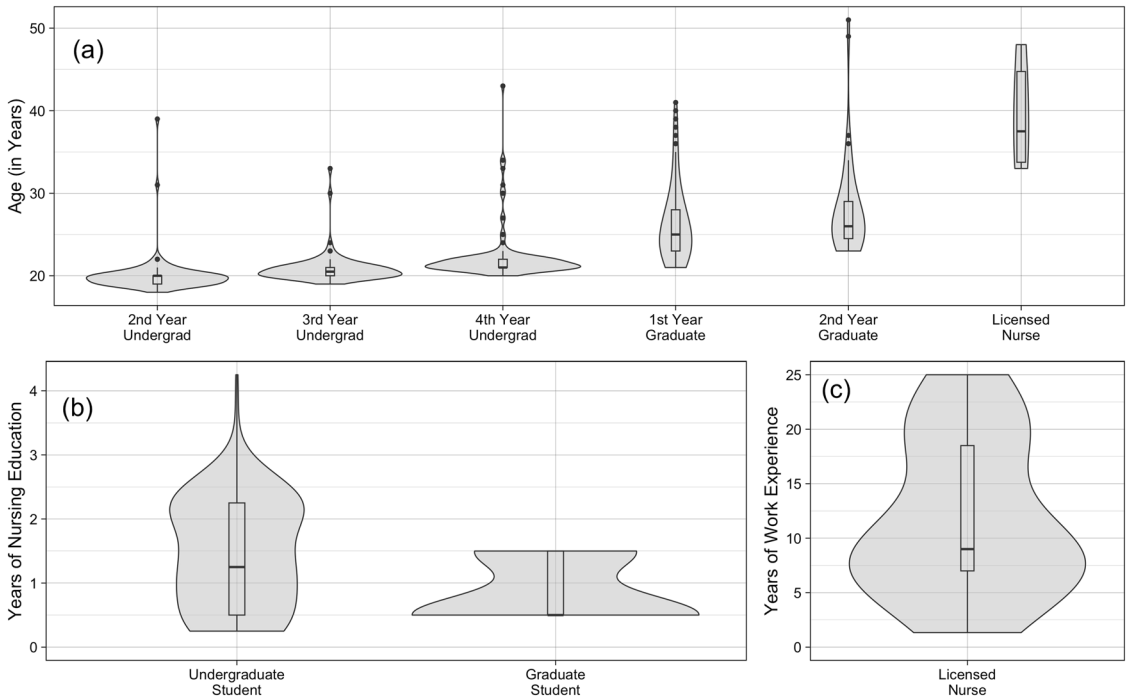

Fig. 3 | Participant cohort demographics. Graphs include the distribution of participants' years of age (**a**), the distribution of students' years of nursing education (**b**), and licensed nurses' years of healthcare experience (**c**). The distribution of participants' years of age is separated by academic program or career (undergraduate student, graduate student, or licensed nurse) and progress (first-year, second-year, third-year, or fourth-year) and shown in order of increasing median age. The distribution of years of nursing curriculum completed is separated by undergraduate and graduate students. These values were estimated by the Assistant Dean of Baccalaureate Nursing Programs based on student course enrollment and pre-licensure program. Years of nursing education included nursing courses only, not general education courses. The distribution of years of healthcare work experience included only licensed nurses. Each distribution is depicted with a violin plot and a boxplot to indicate the relative density (shaded region), median (horizontal line), range (whiskers), interquartile range (box), and outliers (circles). Within each graph, violin plots are scaled so that the area is equivalent across participant groups.

display, or experimental design) corroborating the case and experimental condition completed: 95% of data were verified and the remainder were discarded. Additionally, the time participants spent on each case was checked for adherence to study instructions. Data recorded from interactions with the display indicated participants spent a mean of 2.80 min, a median of 2.47 min, and an interquartile range of 1.71–3.56 min on each case. For inclusion in analysis, participants were permitted 5 min to analyze each case (corresponding to the prediction horizon of the AI algorithm) and 1 additional minute to answer the study questions (which included several additional questions beyond the scope of this analysis that would not be required in clinical practice). Records in which participants spent more than 6 min on a single case were excluded from analysis. If data on the individual timing of cases was not available, records in which participants spent more than 60 min analyzing the series of 10 cases were excluded from analysis. In total, 4125 records were retained for analysis.

To estimate the impacts of AI augmentation on nurses' reported concern, we used a generalized linear mixed effects model. All possible interactions between AI recommendation (0–100%), patient outcome (emergency vs. non-emergency), and experimental condition were included as fixed effects. Random intercepts for each participant were included to account for repeated measures and different concern baselines among participants. Another fully crossed random intercept for each case was included to account for any additional aspects of individual cases that might impact concern beyond what was captured by the AI recommendation and patient outcome[42,84,85]. We normalized concern, centered non-categorical independent variables, and utilized ordered beta regression[86] to model nurses' responses on a continuous closed interval.

From this base model describing our experimental design, we explored several additional independent variables and interactions related to participant experience and potential carry-over effects. First, we included trial number to account for effects of fatigue or learning, although we expected the lack of feedback provided during the study would limit opportunities for the latter. Second, we included the prior experimental condition to account for potential carry-over effects related to switching between conditions. Finally, we explored five variations of participant demographics (three categorical variables and two continuous variables) to account for differences in clinical experience. Categorical variables were based on academic program (undergraduate student, graduate student, and licensed nurse), year in academic program (second-year undergraduate, third-year undergraduate, fourth-year undergraduate, first-year graduate, second-year graduate, and licensed nurse), and years of nursing education (less than 1 year, 1–2 years, more than 2 years, and licensed nurse). Continuous variables included years of experience and the natural logarithm of years of experience[87]. Years of nursing education were estimated by the Assistant Dean of Baccalaureate Nursing Programs based on student course enrollment. Years of experience for licensed nurses was computed as the sum of years of healthcare work experience and years of nursing education. Each of these variables and all possible interactions with variables in the base model were considered as candidate extensions to improve our model. To select among candidate models, we used the Bayesian Information Criterion (BIC) because the purpose of our modeling was to explain rather than to predict[88,89]. The final selected model had 4102 degrees of freedom.

A subset of participants (23%) responded on an integer scale rather than a continuous scale, so we used a multiple imputation technique to estimate the impacts of this unintended interval censorship or grouping error. Both the integer and continuous scale appeared identical to participants, but the former permitted participants to record only integer responses. We assumed participants responded with the integer that was closest to the actual continuous concern they would have reported if they had not been censored. We further assumed participants who were censored by the integer scale would have reported continuous concern in a similar distribution as those who were not censored. Therefore, we

used participants' uncensored responses to construct empirical distributions around each integer (by rounding continuous responses to the nearest integer). We used these empirical distributions to impute interval censored responses. Using multiple imputation with five iterations[90], we compared regression coefficients between the imputed model and non-imputed model[91]. We found no significant differences between any of the models we considered, which is consistent with prior research finding only minor impacts of grouping errors when over 10 categories are utilized[92].

We examined collinearities and residuals to evaluate model fitness prior to performing statistical tests. We used the variance inflation factor (VIF) to check for multicollinearities. We found all VIFs among first-order terms to be less than 5 and all VIFs among interaction terms to be less than 7, suggesting acceptably low correlations among variables (Supplementary Table 2). We also used a simulation-based approach to perform residual diagnostics for hierarchical regression models[93,94]. Model assumptions were determined not to be violated. We utilized this model to perform contrasts between experimental conditions in the cases which corresponded to the most extreme true positive, false positive, true negative, and false negative AI recommendations. Tests were performed on the log odds ratio scale at the level of individual participant responses (i.e., nurses' reports of concern for each patient case). Estimates of means and confidence intervals were back-transformed from the logit scale to the response scale.

Data availability

Data collected from this study are not publicly available because of stipulations in the informed consent process that participant data would not be used or shared for future research. However, descriptive summaries (see Supplementary Table 1) and distributions (see Supplementary Figs. 4–8) of participant data are included in the supplementary materials.

Code availability

Code created for the experimental interface, AI algorithm, and data analysis is available at https://doi.org/10.5281/zenodo.15232639. Patient data visualizations and the logistic regression algorithm utilized in this study were created with R (version 4.2.1) and RStudio (version 2024.04.2) using the library ggplot2 (version 3.5.1) and base package stats. These visualizations were deployed as an online application with the library shiny (version 1.7.2). Further information about the patient data visualizations[35,36] and AI algorithm[31] is published elsewhere. Additional data was collected with online surveys created and deployed with Qualtrics. All data analysis was conducted with R (version 4.2.1) in RStudio (version 2024.04.2) using libraries glmmTMB (version 1.1.9), DHARMa (version 0.4.6), and emmeans (version 1.8.0).

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Author contributions

D.M. and M.R. conceived the study. D.M. planned, developed, and executed data collection and analysis. D.M. and M.R. drafted the manuscript and D.W. contributed to the interpretation of findings. All authors provided critical feedback and substantially contributed to the revision of the manuscript. All authors read and approved the final manuscript.

Competing interests

The authors declare no competing interests.

Ethical considerations

This research complied with all relevant ethical regulations and was approved by the institutional review board at The Ohio State University (identification number 2019H0058). Informed consent was obtained from all participants in this study.

Additional information

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41746-025-01784-y.

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